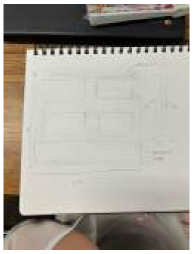
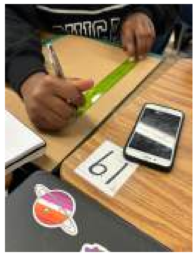
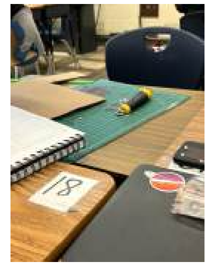
Element G: Construction of a Testable Prototype (Small Scale Prototype)

Day 1 (3/18/2024)


Sketch


Measuring cardboard


Cutting cardboard


Cutting cardboard (continued)


Cutting cardboard (continued)

Description: We began sketching out our design and cut out sized-down cardboard pieces. As shown in the drawing, the dimensions of the testable prototype is 12in x 17in x 2in. Our goal is to make it a horizontal window with features of Design Idea #1 including the planter box and curtains.

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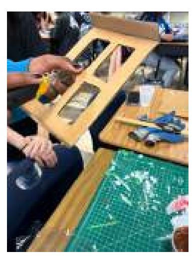
Day 2 (3/19/2024)


Cutting the window


Cutting the window


Cutting the window


Fitting the cardboard


Preparing the lights

Description: We continued cutting the cardboard. We managed to cut window holes using razors and tabs or flaps to stick to each other. We plan on making the distance from the front and back about 2 inches. While working on our prototype, we gathered materials and components to put inside of the space between the front and back. The fairy lights above were collected by one of our group members. We plan on trying to fit it in along with the other layers of the window.

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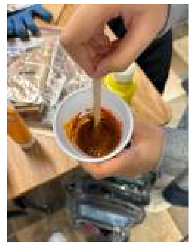
Day 3 (3/20/2024)

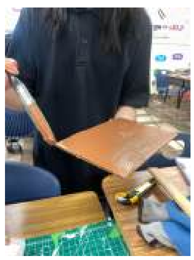

Gathering supplies


Mixing paint


Mixing paint


Fixing window corners


Painting front frame


Painting back frame


Using a fan to dry off the paint quicker

Description: We are finished with cutting the cardboard. We began painting the window a darker and warmer brown color. We plan on adding a layer of paper and lights inside the space of the window later on. Our plan for that is to put the light where the “light source” of the scenery is. We plan on painting the scenery as a sunset or sunrise, according to our customer’s preferences.

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Day 4 (3/21/2024)

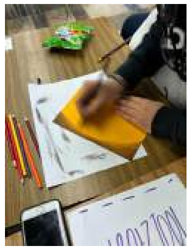

Measuring sides


Adding magnets


Adding magnets to back


Working on lights


Finished lights


Coloring scenery

Description: On Day 4 of prototyping, we attached magnets to the back of our window, set up and glued the overall structure, started working on the background, and attached the lights.

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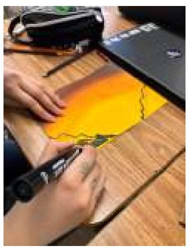
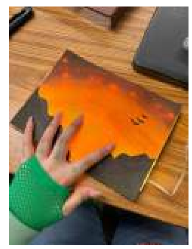
Day 5 (3/22/2024)


Final touches to drawing


Putting everything together


Final drawing with lights at the back


Final touches and extra gluing


Finished product

Description: On Day 5, we finalized our project. First, we finished coloring our drawing of a sunset, leaving space at the bottom for the lights to shine through. The goal is to create an illusion of lights shining or “spreading out,” we tried to achieve this as much as possible but it was limited due to the thickness of the paper and the type of lights we used. We glued the final frames together and attached the lights and other components.

Procedure/Steps

#	Details	Tools and Manufacturing Methods Used
Step 1	Sketch out ideas/designs including size measurements.	🧰 Paper, pencil, ruler ⚙️ Drawing

Step 2	Cut cardboard into 2 pieces, 12in x 17in or less. One will be the front and the other will be the back.	🧰 Paper, pencil, ruler, cardboard, razor blade ⚙️ Cutting
Step 3	Sketch out the window panel design for the front on one of the cardboard.	🧰 Cardboard, paper, pencil, eraser ⚙️ Drawing
Step 4	Cut out the panels and leave enough space in between each panel.	🧰 Scissors, razor blade, ruler, gloves ⚙️ Cutting
Step 5	Create tabs out of cardboard to attach the front and back window parts. These will also be used to determine the space of the components inside the window.	🧰 Pencil, ruler, scissors, razor blade, cardboard, hot glue ⚙️ Drawing
Step 6	Paint the front and back parts of the window. Fix up parts of the design to make it look more clean and neat.	🧰 Acrylic paint, razor blade, hot glue ⚙️ Painting, cutting
Step 7	Add the magnetic components to the back of the window.	Magnets, tape, hot glue 🧰 Paper, pencil, ruler ⚙️ Joining (adhesive bonding)
Step 8	Use a hot glue gun to attach the cardboard pieces together, finally forming the structure of a window.	🧰 Cardboard pieces, hot glue ⚙️ Joining (adhesive bonding)
Step 9	Using the fairy lights, scrunch them up to make the desired image or shape for the light distribution. * Note: our group plans on using a screen light for the final prototype, however, for this test/small-scaled prototype, we will be using fairy lights as a cheaper alternative. The fairy lights we used have spaces in between each bulb, causing the light to look like dots.	🧰 Cardboard window frame, fairy lights, battery box from fairy lights, tape ⚙️ Forming
Step 10	Create the drawing of any scenery. Leave a significant amount of space for the fairy lights to attach and shine through.	🧰 Cardstock paper, oil pastel, drawing mediums ⚙️ Drawing
Step 11	Tape the fairy lights on the paper to the scenery drawing, making sure that all the parts where the light is supposed to shine through align perfectly.	🧰 Scenery drawing, fairy lights, battery box, tape ⚙️ Joining (adhesive bonding)
Step 12	Fix up some corners of the window for good measure (hot glue corners, extra coloring, etc.)	🧰 Markers, hot glue ⚙️ Drawing, joining

Step 13	Put all inside components into the final window.	🧰 Markers, tape, hot glue ⚙️ Joining, finalizing, finishing

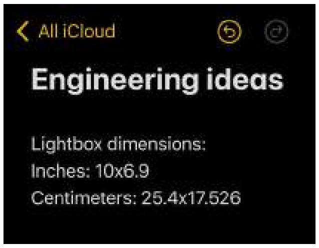
Final Prototype Iteration Explanation (G1)

Material Name	Description	Where Material Was Purchased	Cost per unit	# Units	Total cost per item
Polycarbonate sheet glass	Clear, durable material	Amazon	$20-30	1	$20-30
Wood	Sturdy and reliable material	Home, School	Free	3	$0
A3 Lightbox	Flat and light. plug-in	Amazon	$40-50	1	$40-50
Magnets	Flat circular magnets	Home	Free	4	$0
Dowel	Supports curtains to hang on the window	Home	Free	1	$0
Fabric	Lightweight and decorative material for the window, makes it look more realistic	Home	Free	36inx5in	$0
Tracing Paper	Thin paper, allows light to shine through	Art teacher, Ms. Connealy	Free	3	$0-$10
Markers, colored pencils, oil pastel	Coloring materials	Home	Free	12	$0

Element G (Part 2): Prototype 2.0 Final Product

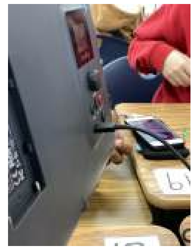
Daily progress

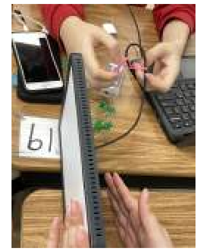
Day 1 (4/9/2024)



Set out dimensions


Mentor's lightbox


Back plug of the lightbox


Side of lightbox


Back functions of the lightbox

<u>Description</u>: On Day 1, we decided to plan out our materials and what we wanted to do with them. An brought the mentor's lightbox to test out the functionality and button features.

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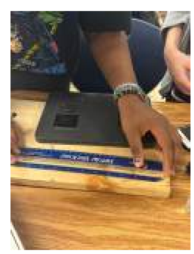
Day 2 (4/11/2024)


New lightbox

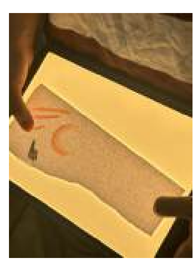

Wood panels


Lightbox unboxing


Measuring the dimensions


Testing different types of paper


Testing colored pencils


Testing the warm light

<u>Description</u>: We gathered materials, discussed wood sizes, bought the lightbox and tested its functions, discussed paper requirements, and tested different paper qualities and coloring materials. We decided on using tracing paper.

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Day 3 (4/12/2024)

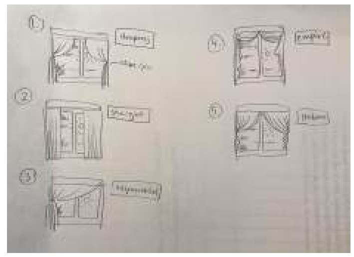

Wood sample


Tracing paper


Discussing wood options


Drawing progress


Dowel


Curtain styles sketch

Description: We gathered paper to draw on, started drawing, got a wood sample (wood collecting in progress), found a dowel for the windows, and started planning curtain designs.

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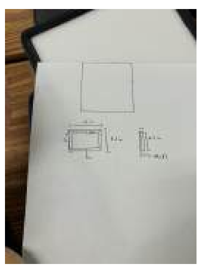
Day 4 (4/15/2024)


Curtains


Sketching measurements


Coloring drawing


Finished curtains


Testing with polycarbonate sheet glass

Description: We worked on the drawing, made the curtains, ironed the curtains, bought polycarbonate sheet glass, and discussed our plans for the front frame.

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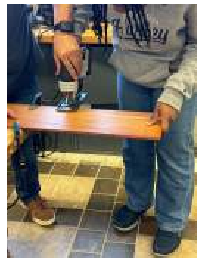
Day 5 (4/16/2024)


Gathering wood

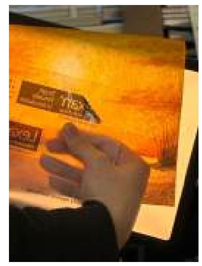

Measuring the wood


Sandiyah cutting wood


Mr. Paz helping with wood cutting


An's TinkerCAD model


Finished drawing


Testing the glass

Description: We finished up drawing, modeled with TinkerCAD, measured wood, drew out wood requirements, and tested the glass and light settings on the lightbox

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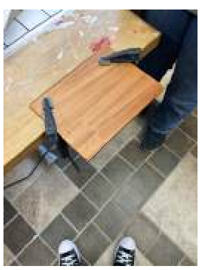
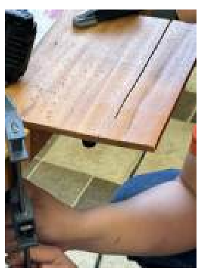
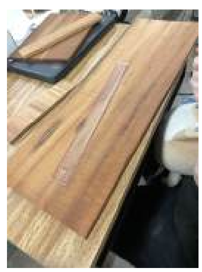
Day 6 (4/17/2024)


Cutting wood


Cutting wood


Measuring the front panel


Measuring the front panel


Sanding down parts


Sanding down parts


Finished sanding down the sides

Description: On Day 6 of prototyping, we continued cutting wood, measuring the front frames and panels, drawing the requirements on the wood. After cutting, we all started sanding down the wood.

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Day 7 (4/18/2024)


Cutting the wood


Cutting the wood


Sanding down the wood


Team Bizzle at work


Testing the glue


Drilling holes in the wood


Sandiyah and Vivian working on their parts

Description: We noticed that our measurements did not satisfy our ideas, so we decided to re-measure and recut our wood.

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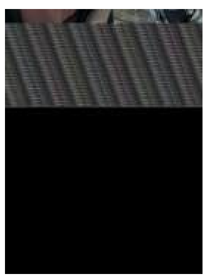
Day 8 (4/19/2024)


Gluing the sides "box"


Re-measuring the front


Working on curtains


Members assisting with glue


Progress on box frame


Had to tape it for stability


Drilling the wood again

Description: We started the gluing process using different types of glue such as wood glue, tacky glue, and hot glue. We practiced drilling holes, in preparation for recutting the frame because some pieces were not straight. We also started working on how we want to assemble the fabric to make the curtains.

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Day 9 (4/22/2024)


Setting up the frame


Preparing to glue


Fixing the drawing


Testing it with the glass


Gluing the frame with wood glue

<u>Description</u>: Our team continued with gluing, securing the area with tape to further ensure that our pieces dry straight. We planned on what we wanted to do next.

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Day 10 (4/23/2024)


Vivian gluing


Sanding down wood


Testing the frame


Measuring the glass


Gluing and taping frame


Recutting the glass


Using bottles to hold stability

Description: We continued gluing, sanded down pieces that we cut, and framed around the lightbox. We also started discussing our plans with the glass and decided that it had to be cut a bit at the top.

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Day 10 (4/24/2024)


Stencil for front frame


Assembling the wood


Testing it in the dark


Assembling the back


Marking wood to cut


Working in progress

Description: We created stencils to ensure the measurements for the front frame were exact, we assembled the wood from the front and back, and cut it during class.

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Day 11 (4/25/2024)


Preparing to cut wood


Sandiyah cutting wood


Testing the light


Sanding down wood

Reusing magnets from previous prototype

Almost done!

<u>Description</u>: We continued cutting the wood pieces, finalized our design, and sanded down parts.

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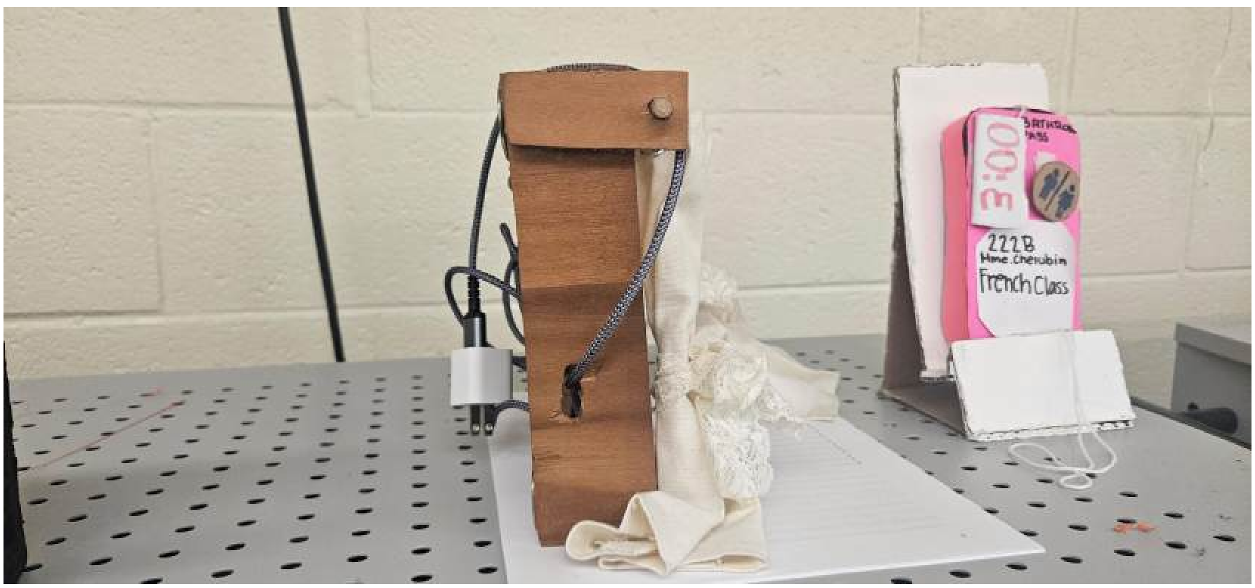
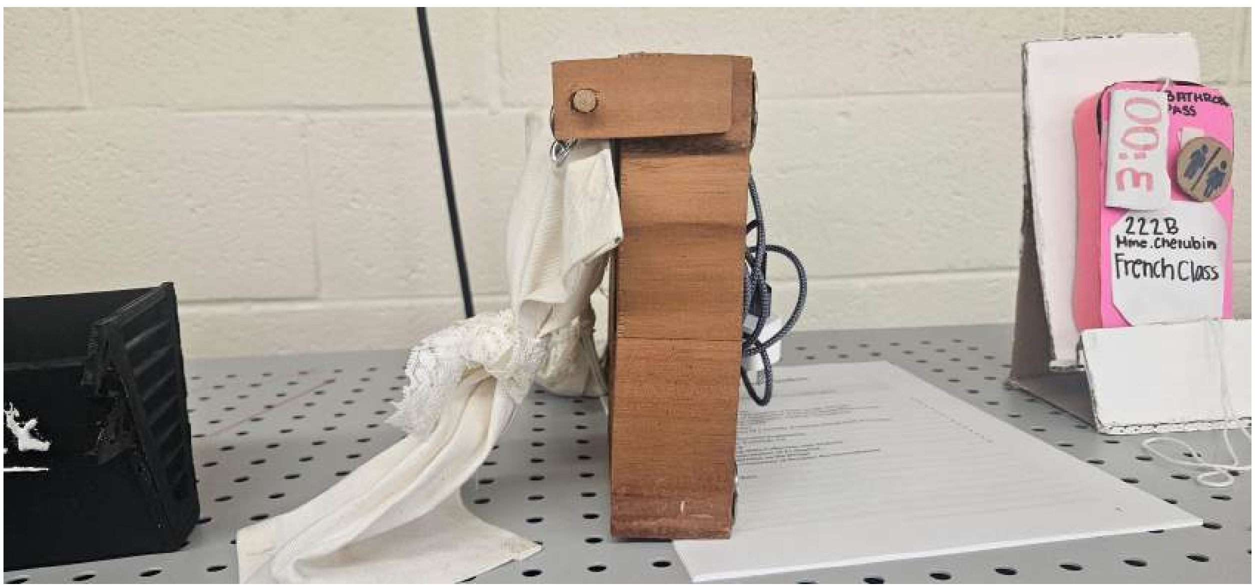
Day 12 (4/26/2024)




3:00
222B
Mme. Cherubin
French Class


3:00
222B
Mme. Cherubin
French Class


SmallRig
3:00
222B


3:00
222B
Mme. Cherubin
French Class

Finished product